

Virtual Reality Training Prototype for Police-Bystander Conflict

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ABSTRACT

The increasing complexity and frequency of societal conflicts pose significant challenges to public safety and demand innovative human-centric policing strategies and solutions. This research presents the development and evaluation of a Virtual Reality (VR) prototype for training Police Officers (POs) to addressing police-offender-bystander conflicts by applying proper conflict de-escalation techniques. The prototype is successfully evaluated by POs considering three scenarios: regular arrest, pandemic rule enforcement, and protest intervention. The results demonstrate the potential of VR solutions to augment traditional training methods by equipping POs with realistic and adaptive strategies to tackle conflict and through this assuring public safety and trust.

Keywords: Virtual Reality, VR, Virtual Reality training, policing, bystanders, police-bystander.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, augmented, and virtual realities; H.5.2 [Information Interfaces and Presentation]: User Interfaces—Interaction Styles.

1 INTRODUCTION

Over the past decade, the number of conflicts that involve Police Officers (POs) significantly increased, driven by a combinations of factors such as public mistrust, social unrest, uncertain societal conditions, and complex community dynamics. These conflicts often escalate due to lack of communication or miscommunication, fast decisions made under pressure, perceived relational gap, sense of closeness, or social distance [1, 2], and a lack of awareness and readiness to diverse (complex) conflict scenarios [3]. To address these challenges, support is seen in recent technological advancements in domains like VR (Virtual Reality), Augmented Reality (AR), and Artificial Intelligence (AI). Accordingly, building responsible and trustworthy technologies in a human-centered approach using VR allows POs to engage with immersive realistic scenarios from where they can practice and learn de-escalation techniques that match the complexities of contemporary conflict [4], enhance their situational awareness, and improve existing communication skills in a safe and controlled environment [5]. Furthermore, developing VR prototypes that rely on scenarios based on real-world events and expert knowledge, fosters trust between law enforcement and the communities they serve, allows the POs to be equipped with effective and ethical methods to handle conflict, and promotes a constructive interaction between police and the public for promoting a safe environment for everyone.

In the context of rising conflicts that involve POs, the role of bystanders is becoming increasingly important. Whether they are just passively witnessing a situation or are actively participating, bystanders have the power to escalate or de-escalate conflicts through their simple presence, actions, and reactions [6]. This power is also influenced by content and experiences that bystanders have and see through digital content which is known to shape public perception and accountability [7]. Moreover, the impact of bystanders in conflict is perceived by POs in time of high conflict as a negative and stressful experience that affects their performance and cognitive capacity [8].

In a VR simulation of a violent conflict during a football match that aims to understand the emerging behavior of bystanders and their cognitive load to decide intervention responses, [9] stresses the fact that sympathy and motivation shape behaviors during stressful situations. [10] builds an adaptive VR game based on Reinforcement Learning (RL) to analyze the behavior of bystanders during an escalating aggressive encounter and assess the aid provided by the bystanders to the victim. With a focus on POs' behavior, [11] address the potential of VR technologies to capture implicit racial bias and understand the perspective of the observer (i.e., bystander) stressing that the more exposure POs have through training, the more they are able to reduce this bias in real-life conflict scenarios. These studies show the potential that VR has for supporting POs in training into conflict de-escalation settings by building or enhancing their de-escalation skills and ability to think critical and manage complex scenarios involving bystanders. Nevertheless, there is a lack of VR solutions that position in a direct setting across various scenarios POs with bystanders. This is important to assure the acquisition of realistic, professional, and ethical skills to deal with the dynamism, uncertainty, and complexities that characterize modern conflicts, and apply effective de-escalation techniques.

To address this gap, this research aims to build a VR prototype for bystander-police conflict for raising awareness and providing training assistance to POs when training on conflict de-escalation methods in a reality mirroring safe environment in a responsible manner. To this end, a human-centered co-design approach is followed by engaging with the POs in all the phases of development of the proposed system and successfully conducting the evaluation process with POs in three different realistic scenarios.

The remainder of this article is structured as follow. Section 2 discusses the design and development of the prototype proposed in this research. Section 3 presents the results of the evaluation process conducted with field experts. Section 4 reflects on the positioning of the prototype proposed in the context of the Mixed Reality (MR) domain. And Section 5 presents concluding remarks and future research perspectives.

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2 PROTOTYPE DEVELOPMENT

Since this research aims to build a VR prototype for police-bystander conflict simulation as a technical artefact with a direct societal contribution, the Design Science Research methodological approach is followed [12] in a co-design human-centered approach [13, 14] under the approval of the ethical assessment conducted by the ethical committee at the Dutch university. This implies executing the following three research phases.

In Phase 1 (Design), a systematic literature review is conducted to explore the use of VR technology in the police domain, existing types of conflicts with police, and complexities of urban conflicts. Studies highlight that conflicts in various (urban) settings frequently escalate due to the unique challenges of high-pressure environments, such as the presence of diverse groups with competing interests and the rapid diffusion of tension among large crowds of people. Additionally, the literature emphasizes the contrasting dynamics of conflicts involving police and perpetrators, both in the absence and presence of bystanders. While without the presence of bystanders, interactions tend to be more controlled, with police relying on direct engagement strategies, in the presence of bystanders the dynamic is altered as their actions, reactions, and involvement can either escalate or de-escalate a situation or conflict. These findings are complemented by a series of four exploratory interviews with Belgian POs with 14.7 years of experience in conflict situations. The experts provided design requirements and advised in the process of building the three scenarios considered in this study, as follows. The first scenario is focused on regular intervention arrest in a relatively peaceful neighborhood during daylight, the second scenario on pandemic rule enforcement to a group of young people in a challenging urban environment, and the third scenario on an intervention made during a protest against the government during evening. In these scenarios four degrees of conflict escalation intensity are considered. Moreover, as examples of de-escalation methods are captured human behavior aspects by allowing POs to introduce themselves on their first name to avoid power differences which is shown to have a 83% success rate, and listen to others by giving space to the people involved to tell their side of the story which is known to have a 79% success rate [15].

In Phase 2 (Development), the solution is implemented in the Unreal Engine platform in C# on a i7 processor laptop with Oculus Quest being used for the testing phase. Here, based on the literature findings and experts' requirements, behavioral trees are built to capture the actions that the agents need to take depending on the conflict escalation intensity for each of the scenarios considered. In this process, the agents observe the scene and use all information available, choose the right set of actions based on these observations, and apply the selected de-escalation methods based on the situation contained in each scenario.

3 EVALUATION

The evaluation process for the VR prototype involved executing three distinct conflict scenarios with five experienced Belgian Police Officers (POs) that have an average of 9.8 years of field experience. The scenarios were designed to simulate a routine arrest, pandemic rule enforcement, and a protest intervention provided a diverse set of conflict settings to evaluate the prototype's capacity to address varying levels of complexity and escalation. Accordingly, each PO engaged individually with the scenarios and applied de-escalation techniques tailored to the situation. Their individual assessments were collected through a structured methodology involving pre- and post-surveys, designed to measure changes in their perceptions, confidence levels, and skill acquisition as a result of the training together with the

effectiveness of the prototype in enhancing situational awareness, decision-making, and communication skills. Additionally, a collective evaluation was conducted by means of a focus group session, where the POs discussed their experiences and provided insights into the realism, usability, and potential impact of the prototype on real-world policing practices. This allowed an in-depth analysis of the system's strengths and improvement areas, providing a way for engaging with collective feedback and recommendations.

Specifically, the Oculus Quest 2 VR headset was used in a safe room for assuring a lifelike experience. A capture of the initial scene of the third scenario is depicted in Figure 1.

In this process, the evaluation metrics used included feasibility, ease of use, stability, and acceptability, each evaluated on a five-point Likert scale ranging from 1 (very poor) to 5 (excellent). These metrics were selected to capture both the technical and experiential aspects of the system for assuring a comprehensive assessment of its suitability for real-world application. Feasibility, which measures the practicality and implementation potential of the VR system, received an average score of 3.6, indicating moderate success with room for improvement in optimizing operational workflows. Ease of use, reflecting the intuitiveness and user-friendliness of the interface, achieved a high average score of 4.4, demonstrating that the system is accessible to users with varying levels of technical expertise. Stability, which evaluates the system's reliability and resistance to crashes or errors, scored 4.0, confirming a robust technical performance with minimal disruptions during training. Finally, acceptability, which gauges the overall satisfaction and perceived value of the system by the officers, recorded an average of 4.2, suggesting strong approval and alignment with user needs.



Figure 1: Capture Scenario 3 (protest intervention).

These results show that the prototype proposed has a solid foundation for further integration with existing police programs and problems that could be addressed (feasibility), has an intuitive and user-friendly design, facts that are critical for technological adoption (ease of use), shows the importance of enhancing reliability to maintain realism and immersion during training (stability), and reflects the fact that the POs see it as an innovative, relevant, and beneficial tool for improving situational awareness and de-escalation skills (acceptability).

As addition from the collective assessment, the experts consider that the prototype is a supportive method for POs and would be beneficial to use it for real school police training as it would allow them to build valuable conflict de-escalation skills in a safe immersive environment.

4 POSITIONING IN THE MIXED REALITY CONTEXT

The prototype proposed could be further implemented in a MR context in order to enhance its realism, usability, and the overall impact by considering a blend between virtual technologies with real-world mechanisms and elements. As the VR solution is

simulated on three different case scenarios, each featuring escalating conflict intensities and incorporating evidence-based de-escalation techniques, transitioning this solution into a MR environment would involve integrating aspects such as physical props like batons and handcuffs, that are synchronized with the virtual simulation for providing haptic feedback together with more advanced MR headsets to be able to overlay virtual environments onto physical spaces. Accordingly, the MR system would operate through a hybrid interaction model, where virtual avatars, programmed with context-sensitive behavioural trees, simulate bystander dynamics and offender reactions while dynamically adapting to the POs' real-time verbal and physical responses tracked through headsets and auxiliary sensors. Moreover, MR extends the solution's applicability by allowing POs to train within familiar environments that reduce logistical constraints and improve transferability to field conditions while supporting teamwork and reliable coordinated response strategies.

5 CONCLUSION

This research presents the development of a VR prototype for POs training that addresses building skills and decision-making support for using de-escalation methods in conflicts in the presence of bystanders, an important societal issue that needs more attention by both academic and practitioner communities. Being developed following the Design Science Research methodology and adopting a co-design stance in a human-centered approach, the prototype was successfully demonstrated and evaluated considering three different case scenarios by Belgian POs. The experts acknowledge its value, effectiveness, and utility for integration into police academy training programs, and further development. Therefore, this research lays a foundation for further research and development in two directions. First, by considering the integration of more dimensions of conflict and human factors (e.g., psychological and cognitive) that characterize police conflict with bystanders. And second, by considering the integration of various AI (Artificial Intelligence) techniques that would allow building intelligent and adaptive mechanisms for tackling and promoting resolution of police conflict with bystanders in a multimodal approach.

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